

Midterm 1 — Solutions

M 375T Fall 2026 — 100 points. Q1–Q4: 25 pts each.

1. Ordinary and lazy walks, in the long run.

- (a) The path 1–0–2 (vertex 0 in the middle), all weights 1. $d = (2, 1, 1)$; row 0 of P is $(0, \frac{1}{2}, \frac{1}{2})$ and rows 1, 2 are $(1, 0, 0)$. Eigenvalues of P : 1, -1 , 0. Two routes: (i) $\det(P - \lambda I) = -\lambda^3 + \lambda = -\lambda(\lambda - 1)(\lambda + 1)$; (ii) P and $P^\top$ have the same eigenvalues, and with $(P^\top x)_0 = x_1 + x_2$, $(P^\top x)_j = \frac{1}{2}x_0$ one gets $P^\top w_1 = w_1$, $P^\top w_2 = (-2, 1, 1) = -w_2$, $P^\top w_3 = 0$. (Eigenvectors of P itself: $(1, 1, 1)$, $(1, -1, -1)$, $(0, 1, -1)$.) 7 pts: 4 drawing + $d + P$, 3 for the three eigenvalues (1 each); either route.
- (b) Write $\mu_0 = e_1 = a_1 w_1 + a_2 w_2 + a_3 w_3$. Entry 0: $2a_1 + 2a_2 = 0$; entry 2: $a_1 - a_2 - a_3 = 0$; entry 1: $a_1 - a_2 + a_3 = 1$. Hence $a_2 = -a_1$, $a_3 = 2a_1 = \frac{1}{2}$, $a_1 = \frac{1}{4}$: $\mu_0 = \frac{1}{4}w_1 - \frac{1}{4}w_2 + \frac{1}{2}w_3$. Then $\mu_t = (P^\top)^t \mu_0 = \frac{1}{4}w_1 - \frac{1}{4}(-1)^t w_2 + 0^t \cdot \frac{1}{2}w_3$, i.e. for $t \geq 1$: $\mu_t = \frac{1}{4}(w_1 + w_2) = (1, 0, 0)$ for odd t and $\mu_t = \frac{1}{4}(w_1 - w_2) = (0, \frac{1}{2}, \frac{1}{2})$ for even t . Check: $\mu_1 = \text{row 1 of } P = e_0$, $\mu_2 = \text{row 0} = (0, \frac{1}{2}, \frac{1}{2})$, $\mu_3 = e_0$. 100 is even: line 1 is $[0. \quad 0.5 \quad 0.5]$. The law does not converge: the w_2 term has $|-1|^t = 1$ and flips sign every step (the graph is bipartite, $\{0\}$ versus $\{1, 2\}$). 10 pts: 3 coefficients, 3 formula for μ_t , 4 for line 1. Any correct route earns full credit (e.g. computing μ_1, μ_2, μ_3 , spotting the period-2 pattern, and arguing that 100 is even). A bare answer with no work: at most 5.
- (c) $Q^\top = \frac{1}{2}(I + P^\top)$, so the eigenvalues become $\frac{1}{2}(1 + \lambda)$: 1, 0, $\frac{1}{2}$. $\nu_t = \frac{1}{4}w_1 - \frac{1}{4} \cdot 0^t w_2 + (\frac{1}{2})^t \cdot \frac{1}{2}w_3 \rightarrow \frac{1}{4}w_1 = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$; at $t = 100$ the $(\frac{1}{2})^{100}$ term is invisible after rounding. Line 2: $[0.5 \quad 0.25 \quad 0.25]$. The limit is the invariant measure $\pi = d/4$. It converges because every eigenvalue of $Q^\top$ other than 1 has absolute value < 1 : laziness moved the troublesome -1 to 0, so the oscillating mode dies after one step. 8 pts: 3 eigenvalues of $Q^\top$, 2 for the decaying terms / formula, 3 for line 2. “Converges to the invariant measure $\pi = d/4$ ” with a reason is a full-credit route. Uniform $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$: at most 3.

[25 pts]

2. The best Gaussian, mean and variance.

- (a) $H(m, s^2) = \frac{1}{n} \sum_{i=1}^n \left[\frac{1}{2} \log(2\pi s^2) + \frac{(x_i - m)^2}{2s^2} \right] = -\frac{1}{n} \sum_i \log p_{m, s^2}(x_i)$, the **cross-entropy** of the data against $N(m, s^2)$ (negative log-likelihood per point). 5 pts for the formula (the name is not required).
- (b) $\bar{x} = \frac{1+2+6+7}{4} = 4$, $\hat{\sigma}^2 = \frac{9+4+4+9}{4} = 6.5$. Line 1: 4.0 6.5. 3 pts.
- (c) $\partial H / \partial m = -\frac{1}{s^2} \cdot \frac{1}{n} \sum_i (x_i - m) = -\frac{\bar{x} - m}{s^2}$, zero exactly at $m = \bar{x}$; H is a convex parabola in m , so $\bar{x}$ is the minimizer for every s^2 . Hence $H(4, 6.5) < H(3, 6.5)$: line 3 is **True**. 7 pts: 3 derivative, 2 minimizer $\bar{x}$ with convexity, 2 for **True**.
- (d) With $m = \bar{x}$: $H(\bar{x}, s^2) = \frac{1}{2} \log(2\pi s^2) + \frac{\hat{\sigma}^2}{2s^2}$; $\partial H / \partial s^2 = \frac{1}{2s^2} - \frac{\hat{\sigma}^2}{2s^4} = 0 \iff s^2 = \hat{\sigma}^2$. $H \rightarrow \infty$ as $s^2 \rightarrow 0$ and as $s^2 \rightarrow \infty$, so this critical point is the minimum. Hence $H(4, 6.5) < H(4, 4)$: line 2 is **True**. 10 pts: 3 derivative, 5 minimizer $\hat{\sigma}^2$ with the endpoint argument, 2 for **True**.

[25 pts]

3. The energy of a cut.

- (a) A 4-cycle $0-1-2-3-0$ with weights $W_{01} = 3$, $W_{12} = 1$, $W_{23} = 2$, $W_{30} = 1$. $d = (4, 4, 3, 3)$

and $L = D - W = \begin{pmatrix} 4 & -3 & 0 & -1 \\ -3 & 4 & -1 & 0 \\ 0 & -1 & 3 & -2 \\ -1 & 0 & -2 & 3 \end{pmatrix}$. 5 pts: 2 drawing + d , 3 L (rows sum to zero).

- (b) Only edges joining a $+1$ to a -1 contribute, each $W_{ij}(2)^2 = 4W_{ij}$: edge $1-2$ gives 4 and edge $3-0$ gives 4; edges $0-1$ and $2-3$ join equal values and give 0. Line 1: 8.0. 5 pts: 3 for the two edges, 2 for 8.
- (c) $(Lv)_i = d_i v_i - \sum_j W_{ij} v_j$: at 0: $4 - (3 - 1) = 2$; at 1: $4 - (3 - 1) = 2$; at 2: $-3 - (1 - 2) = -2$; at 3: $-3 - (-2 + 1) = -2$. Line 2: $[2. \quad 2. \quad -2. \quad -2.] = 2v$, so v is an eigenvector of L with eigenvalue 2. 8 pts: 5 for Lv , 3 for eigenvector with $\lambda = 2$.
- (d) $\text{cut}(A, B) = W_{12} + W_{30} = 2$; $\text{vol}(A) = 4 + 4 = 8$, $\text{vol}(B) = 3 + 3 = 6$; $N_{\text{cut}} = 2(\frac{1}{8} + \frac{1}{6}) = \frac{7}{12}$. Since $v_i - v_j = \pm 2$ across every cut edge and 0 otherwise, $v^T L v = 4 \text{cut}(A, B)$: $8 = 4 \cdot 2$. 7 pts: 5 for $\text{cut}/\text{vol}/N_{\text{cut}}$, 2 for $v^T L v = 4 \text{cut}$.

[25 pts]

4. Catching the impostor.

- (a) `(50000,): sum(axis=1)` collapses the 400 columns, one number per row; an entry of k counts how many of the 400 uniform samples in that data set fall in $(-c, c)$. 5 pts: 3 shape, 2 sentence.
- (b) The uniform on $[-\sqrt{3}, \sqrt{3}]$ has density $\frac{1}{2\sqrt{3}}$, so $\mathbb{P}(|X| < c) = \frac{2c}{2\sqrt{3}} = \frac{c}{\sqrt{3}} = \frac{0.6745}{1.732} \approx 0.389$. Line 2 is close to 0.389 (law of large numbers over the 50,000 rows), not to the model's $\frac{1}{2}$. 8 pts: 2 for the density, 3 for the length ratio $c/\sqrt{3}$, 3 for 0.389.
- (c) $k \sim \text{Binomial}(400, q)$ with $q \approx 0.389$: variance $400 \cdot 0.389 \cdot 0.611 \approx 95$, standard deviation ≈ 9.8 . Line 3 is close to 9.8. 5 pts: 2 for $\text{Binomial}(400, q)$, 3 for ≈ 9.8 (accept 9.7–9.8; 10 with $q = \frac{1}{2}$ earns 1).
- (d) For genuine $N(0, 1)$ rows, $2nD \approx \chi_1^2$ and 3.84 is its 95% quantile, so line 4 would print ≈ 0.05 . Here a typical row has $\varepsilon \approx 0.389 - 0.5 = -0.111$, so $D \approx 2\varepsilon^2 \approx 0.0246$ and $2nD \approx 800 \cdot 0.0246 \approx 19.7 \gg 3.84$: almost every impostor data set is rejected, and line 4 is close to 1 (about 0.99). This is Lecture 5's "how many samples until we know": at $n = 400$ the two-bin test detects the impostor with probability about 99%. 7 pts: 3 for $\varepsilon \approx -0.11$ and $2nD \approx 20 \gg 3.84$, 2 for the answer ≈ 1 , 2 for the contrast with 0.05 under the model (χ_1^2). Answering 0.05: at most 2.

[25 pts]

Coverage. None of the four questions appears on the practice sheet; each uses a graph or a computation the sheet treats in a non-code exercise. Q1: the three-vertex path (the star $K_{1,2}$) in the spirit of practice D3 and E1, but via the eigenvector expansion of practice F2–F3 (eigenvectors of P^T given; $\mu_t = \sum a_k \lambda_k^t w_k$; -1 versus 0); Lecture 8 §3, §6. Q2: the joint minimizer of practice B2(d) (Lecture 3 §5, Lecture 4 §2), tested by comparing H at two points instead of a grid. Q3: the weighted 4-cycle of Quiz 4, Quiz 5 Q1 and practice D4/F2 with new weights 3, 1, 2, 1 (Lecture 8 §8, Lecture 9 §3–§4, Lecture 10 §1). Q4: Lecture 5 §3–§4 (the impostor $U[-\sqrt{3}, \sqrt{3}]$, $q_1^* = c/\sqrt{3} = 0.389$, power 99% at $n = 400$) in the code form of practice C3 and Practice 4 A5. Expected discriminators: Q1(b) (the coefficients a_k and the $(-1)^t$ term), Q2(d) (the s^2 derivative), Q4(b) (the length ratio, not $\frac{1}{2}$).